

# Comprehensive Road Scene Understanding for Autonomous Driving: From Mask-Based Segmentation to Anomaly Detection

Farhad Mahsa Aein Kamyar

Politecnico di Torino — Artificial Intelligence Course Project

## Abstract

Deep segmentation networks achieve strong closed-set performance but fail on unknown, out-of-distribution (OoD) objects, a critical safety gap for autonomous driving. In this project we study semantic, instance and panoptic segmentation, compare two pre-trained *Encoder-only Mask Transformer* (EoMT) checkpoints—one trained on Cityscapes and one on COCO—under a single consistent evaluation pipeline, fine-tune the COCO model towards the road domain, and build a suite of post-hoc anomaly-segmentation baselines on top of both a pixel-based network (ERFNet) and the mask-based EoMT. We evaluate four scoring rules (MSP, Max Logit, Max Entropy and Rejection-by-All) plus temperature scaling on five anomaly benchmarks using AuPRC and FPR<sub>95</sub>. Our results show that the mask architecture vastly outperforms the pixel baseline (e.g. 38 → 77 AuPRC on RoadAnomaly-21), that closed-set domain knowledge directly drives open-set detection (the off-domain COCO model is near-useless until fine-tuned), and that the strongest scoring rule depends on how well calibrated the underlying classifier is. Source code: <https://github.com/<YOUR-GROUP-REPO>>.

## 1 Introduction

**Project context.** Modern perception systems rely on dense prediction tasks. *Semantic segmentation* assigns a class to every pixel, *instance segmentation* separates individual objects of the same class, and *panoptic segmentation* [11] unifies the two. While these tasks are largely solved in well-defined, closed-world settings, deployed networks behave unpredictably on *anomalous* or *out-of-distribution* (OoD) objects that were never seen during training. For safety-critical applications such as autonomous driving, an unexpected object on the road (lost cargo, animals, debris) must be flagged rather than silently mis-classified as a known class. The goal of this project is therefore twofold: (i) understand and compare mask-based segmentation architectures on known classes, and (ii) design and evaluate an *anomaly-segmentation* pipeline that identifies unknown regions in road scenes.

**Literature foundations.** Real-time pixel-wise segmentation is exemplified by ERFNet [10], an efficient residual factorized convolutional network. A paradigm

shift came with *mask classification*: MaskFormer [3] showed that predicting a set of binary masks, each with a single class label, generalises per-pixel classification, and Mask2Former [4] unified semantic, instance and panoptic segmentation with masked attention. Recently, EoMT [6] demonstrated that a plain Vision Transformer, initialised with the self-supervised features of DINOv2 [5], is “secretly” a segmentation model: it appends learnable object queries directly to the ViT and needs no specialised decoder. For anomaly detection we rely on *post-hoc* confidence measures: Maximum Softmax Probability [14], the Max Logit score from scaling OoD detection to real-world settings [8], the entropy of the predictive distribution, and—specific to mask architectures—Rejection-by-All (RbA) [7], which flags pixels that *no* known-class query claims. We evaluate on the Segment-MeIfYouCan [1] and Fishyscapes [2] benchmarks, using Cityscapes [12] and COCO [13] as in-distribution data, and explore parameter-efficient adaptation inspired by LoRA [9].

## 2 Methodology

**Models.** We use two architectures. **ERFNet** is a compact encoder-decoder CNN that outputs per-pixel class logits over the 19 Cityscapes training classes. **EoMT** couples a DINOv2 ViT-Base/14 backbone with  $N$  learnable queries; its forward pass returns, per query, a class distribution and a binary mask. We work with three EoMT checkpoints: one trained on Cityscapes (semantic,  $N=100$ , 19 classes), one trained on COCO (panoptic,  $N=200$ , 133 classes), and the fine-tuned model produced in this work.

**From masks to pixels.** For any scoring rule we collapse the mask outputs into a per-pixel class score map

$$L_c(p) = \sum_{q=1}^N \sigma(m_q(p)) \text{softmax}(z_q)_c, \quad (1)$$

where  $m_q$  are the mask logits and  $z_q$  the class logits of query  $q$ , and the “no-object” class is dropped. The argmax of  $L$  gives the semantic prediction.

**Post-hoc anomaly scores.** Given per-pixel logits  $\ell_c(p)$  (raw logits for ERFNet,  $L_c$  for EoMT) we compute, per pixel, the anomaly score  $s(p)$  as: MSP  $s = 1 - \max_c \text{softmax}(\ell)_c$ ; Max Logit  $s = -\max_c \ell_c$ ; Max Entropy  $s = -\sum_c p_c \log p_c$  with  $p = \text{softmax}(\ell)$ ; and, for the mask models only, RbA  $s = -\sum_c L_c(p)$ , i.e. a pixel is anomalous when the total response across all known classes is low. *Temperature scaling* replaces  $\text{softmax}(\ell)$  by  $\text{softmax}(\ell/T)$  before the MSP step.

**Consistent cross-dataset evaluation.** Comparing the COCO and Cityscapes models is non-trivial because their label spaces differ (133 vs. 19 classes). We therefore map COCO predictions onto the Cityscapes label space using the official category identifiers and evaluate *only* the 16 classes that genuinely overlap, excluding pole, traffic sign and rider (no COCO equivalent). The identical pipeline—letterbox resize to the model’s native square input, single forward pass, evaluation at  $512 \times 1024$ —is applied to every model so that mIoU differences reflect the models, not the protocol.

**Fine-tuning.** To adapt the COCO model to the road domain under a tight compute budget we follow the lightweight strategy suggested in the project brief: the DINOv2 backbone, the queries and the mask head are *frozen*, and only the classification head is replaced by a fresh 19-class layer and trained. We use a per-pixel cross-entropy on the semantic score map (a simplified objective rather than the full Hungarian mask-matching loss), AdamW, automatic mixed precision, and a 1000-image subset of the Cityscapes training set for two epochs.

**Metrics.** Closed-set quality is measured with mean Intersection-over-Union (mIoU). Anomaly quality uses Area under the Precision-Recall Curve (AuPRC, higher is better) and the false-positive rate at 95% true-positive rate ( $\text{FPR}_{95}$ , lower is better), computed over all valid pixels of each benchmark.

### 3 Experimental Results

**Closed-set comparison (Step 4–5).** Table 1 reports per-class mIoU on the 16 shared classes of the Cityscapes validation set. As a sanity check, ERFNet reaches 72.2 mIoU over all 19 classes, matching its published performance and validating our pipeline.

**Anomaly segmentation baselines (Step 7–8).** Table 2 reports the full set of post-hoc baselines for the pixel model (ERFNet) and the three EoMT checkpoints across the five benchmarks.

**Temperature scaling.** Table 3 sweeps the temperature  $T$  for the MSP score of the Cityscapes EoMT model.

**Table 1:** Per-class IoU and mIoU (%) on the 16 shared classes of Cityscapes val. FT = COCO model after head fine-tuning.

| Class            | City-EoMT   | COCO-EoMT   | COCO→FT     |
|------------------|-------------|-------------|-------------|
| road             | 98.1        | 93.4        | 96.3        |
| sidewalk         | 84.8        | 61.3        | 72.3        |
| building         | 92.2        | 83.1        | 86.5        |
| wall             | 63.8        | 47.9        | 48.5        |
| fence            | 61.8        | 39.2        | 43.5        |
| traf. light      | 65.9        | 45.8        | 46.7        |
| vegetation       | 91.0        | 84.3        | 85.5        |
| terrain          | 62.8        | 53.6        | 56.5        |
| sky              | 93.9        | 87.0        | 89.3        |
| person           | 78.1        | 60.0        | 63.2        |
| car              | 93.8        | 80.7        | 89.9        |
| truck            | 78.5        | 27.3        | 64.7        |
| bus              | 87.8        | 64.8        | 69.9        |
| train            | 77.8        | 4.2         | 28.3        |
| motorcycle       | 64.4        | 51.7        | 52.6        |
| bicycle          | 74.3        | 61.9        | 62.1        |
| <b>mIoU (16)</b> | <b>79.3</b> | <b>59.2</b> | <b>66.0</b> |

**Quantitative breakdown.** The mask architecture improves anomaly AuPRC over the pixel baseline by a wide margin on every benchmark (e.g. RA-21  $38 \rightarrow 77$ , Road Anomaly  $16 \rightarrow 78$ ). Among scoring rules, Max Logit is best for ERFNet, while Max Entropy is marginally best for EoMT-Cityscapes. The COCO model is essentially unusable for road anomalies (RO-21 AuPRC 1.4,  $\text{FPR}_{95} \approx 100$ ). Fine-tuning lifts the COCO model’s mIoU from 59.2 to 66.0 and, crucially, its anomaly AuPRC on several benchmarks (RO-21  $1.4 \rightarrow 65$ , FS Static  $4 \rightarrow 56$ , Road Anomaly  $20 \rightarrow 47$ ). Temperature scaling changes AuPRC by  $< 0.3$  points.

## 4 Discussion

**Why mask models win.** EoMT’s advantage stems from two factors. First, its DINOv2 backbone provides rich, self-supervised features that generalise beyond the training label set, so unknown objects produce distinctive responses. Second, mask classification decouples “where” (the masks) from “what” (the class), so an unknown object can occupy a region that no class query confidently claims—precisely the signal RbA exploits. The pixel-based ERFNet, by contrast, must assign every pixel to one of 19 classes and is forced into over-confident, locally smooth predictions, leaving little usable uncertainty.

**Closed-set knowledge enables open-set detection.** The most important finding is the tight link between mIoU and anomaly quality. The COCO model has never learned what a “normal” road looks like, so it cannot recognise what deviates from it; its anomaly scores are near-random. After fine-tuning only a 15k-parameter head on Cityscapes, both its mIoU and its anomaly de-

**Table 2:** Anomaly segmentation results: AuPRC ( $\uparrow$ ) / FPR<sub>95</sub> ( $\downarrow$ ), in %. mIoU is the closed-set quality of each model (constant across scoring rules). Best AuPRC per benchmark in **bold**.

| Model     | mIoU | Method      | SMIYC RA-21  |       | SMIYC RO-21  |       | FS L&F       |       | FS Static    |       | Road Anomaly |       |
|-----------|------|-------------|--------------|-------|--------------|-------|--------------|-------|--------------|-------|--------------|-------|
|           |      |             | AuPRC        | FPR   | AuPRC        | FPR   | AuPRC        | FPR   | AuPRC        | FPR   | AuPRC        | FPR   |
| ERFNet    | 72.2 | MSP         | 29.10        | 62.55 | 2.71         | 65.22 | 1.73         | 50.56 | 5.03         | 40.90 | 12.42        | 82.58 |
|           |      | Max Logit   | <b>38.32</b> | 59.34 | <b>4.63</b>  | 48.44 | <b>3.28</b>  | 45.45 | <b>6.27</b>  | 39.65 | <b>15.58</b> | 73.25 |
|           |      | Max Entropy | 30.97        | 62.66 | 3.04         | 65.91 | 2.56         | 50.13 | 5.89         | 40.64 | 12.67        | 82.75 |
| EoMT-City | 79.3 | MSP         | 74.26        | 33.43 | <b>90.19</b> | 0.53  | 24.64        | 15.35 | 29.90        | 34.47 | 75.76        | 19.37 |
|           |      | Max Logit   | 73.71        | 33.55 | 90.14        | 0.56  | 24.67        | 15.10 | 30.12        | 34.15 | 75.13        | 19.12 |
|           |      | Max Entropy | <b>76.68</b> | 33.30 | 90.12        | 0.67  | <b>27.47</b> | 15.17 | 28.63        | 34.38 | <b>77.65</b> | 18.86 |
|           |      | RbA         | 72.82        | 45.64 | 88.64        | 69.31 | 25.13        | 11.23 | <b>32.59</b> | 34.67 | 75.03        | 17.31 |
| EoMT-COCO | 59.2 | MSP         | 34.36        | 87.97 | 1.43         | 99.99 | 1.42         | 96.73 | 4.14         | 97.13 | 20.74        | 94.05 |
|           |      | Max Logit   | 34.96        | 87.78 | 1.42         | 99.99 | 1.41         | 96.74 | 4.11         | 97.21 | 21.00        | 94.22 |
|           |      | Max Entropy | <b>41.01</b> | 80.14 | 3.72         | 99.99 | 1.14         | 96.52 | 3.88         | 97.94 | <b>23.39</b> | 92.74 |
|           |      | RbA         | 37.45        | 62.24 | <b>36.77</b> | 100.0 | 0.28         | 96.65 | 1.27         | 99.95 | 20.65        | 84.38 |
| EoMT-FT   | 66.0 | MSP         | <b>44.18</b> | 43.22 | <b>65.48</b> | 99.98 | <b>0.32</b>  | 92.34 | <b>55.95</b> | 48.88 | <b>47.28</b> | 85.19 |
|           |      | Max Logit   | 40.96        | 44.55 | 62.53        | 99.99 | 0.30         | 94.66 | 53.83        | 48.63 | 45.45        | 86.59 |
|           |      | Max Entropy | 37.90        | 44.93 | 62.77        | 99.99 | 0.29         | 93.30 | 53.52        | 48.22 | 44.02        | 89.91 |
|           |      | RbA         | 15.72        | 89.19 | 23.53        | 99.99 | 0.17         | 99.13 | 32.75        | 74.83 | 25.32        | 96.86 |

**Table 3:** Temperature scaling of MSP (EoMT-Cityscapes), AuPRC (%). fair way to handle such mismatches.

| $T$  | RA-21        | RO-21 | FS L&F       | FS Static    | R.Anom       |
|------|--------------|-------|--------------|--------------|--------------|
| 0.50 | 74.14        | 90.18 | 24.63        | 29.87        | 75.60        |
| 0.75 | 74.20        | 90.18 | 24.64        | 29.89        | 75.69        |
| 1.00 | 74.26        | 90.19 | 24.64        | 29.90        | 75.76        |
| 1.10 | 74.27        | 90.19 | 24.65        | 29.90        | 75.78        |
| 2.00 | <b>74.39</b> | 90.19 | <b>24.67</b> | <b>29.93</b> | <b>75.92</b> |

tection improve substantially. This confirms that OoD detection is fundamentally relative to a well-modelled in-distribution.

**Scoring-rule behaviour and the RbA caveat.** RbA is the strongest rule on some benchmarks for the well-trained Cityscapes model, but it *collapses* on the fine-tuned model (RA-21 AuPRC 15.7 vs. MSP’s 44.2). RbA sums un-normalised class responses and therefore depends on the *magnitude/calibration* of the classifier, which our lightweight cross-entropy fine-tuning did not preserve. MSP, Max Logit and Max Entropy, which read the shape or the maximum of the distribution, are far more robust to this recalibration. A practical lesson is that score choice should follow the calibration state of the model. The flatness of temperature scaling has the same root cause:  $L$  is already a product of sigmoid and softmax terms, so a second temperature has little leverage.

**Cross-dataset class mismatch.** The COCO→Cityscapes comparison exposes genuine semantic mismatches. The clearest is the **train** class: Cityscapes “train” denotes urban trams, whereas COCO “train” denotes mainline locomotives, yielding only 4.2 mIoU for the COCO model versus 77.8 for the native one. Restricting the comparison to shared concepts (and reporting this case explicitly rather than hiding it) is the

**Methodological critique / limitations.** Several simplifications bound our absolute numbers. (i) We use a single letterboxed forward pass instead of EoMT’s sliding-window inference, so absolute mIoU is below the official figures—though the protocol is identical for all models, preserving fairness. (ii) Fine-tuning trains only the classification head with a per-pixel cross-entropy, not the full Hungarian mask-matching objective, and on a 1000-image subset; the frozen COCO masks cap the achievable mIoU (66.0 vs. 79.3). (iii) Our RbA follows the summed-softmax variant; the original multi-label formulation may behave differently. (iv) FS Lost&Found remains hard for all COCO-derived models because its anomalies are small and distant. (v) FPR<sub>95</sub> is occasionally near 100% even when AuPRC is decent, indicating that the score ranking is informative but the operating threshold is poorly calibrated—a clear direction for future calibration work (e.g. LoRA-based fuller fine-tuning).

## 5 Conclusion

We built and evaluated a complete road-scene anomaly-segmentation pipeline, comparing a pixel-based network with a state-of-the-art mask transformer across four post-hoc scoring rules and three checkpoints. Mask-based EoMT dramatically outperforms ERFNet, closed-set domain knowledge is shown to be the key driver of open-set detection, and the best scoring rule depends on model calibration. A lightweight head-only fine-tuning recovers most of the COCO model’s domain gap at negligible cost, while highlighting that score calibration—not just accuracy—governs reliable anomaly detection.

## References

- [1] R. Chan *et al.*, “SegmentMeIfYouCan: A Benchmark for Anomaly Segmentation,” *NeurIPS Datasets and Benchmarks*, 2021.
- [2] H. Blum *et al.*, “The Fishyscapes Benchmark: Anomaly Detection for Semantic Segmentation,” *IJCV*, 2021.
- [3] B. Cheng, A. Schwing, A. Kirillov, “Per-Pixel Classification is Not All You Need for Semantic Segmentation,” *NeurIPS*, 2021.
- [4] B. Cheng *et al.*, “Masked-attention Mask Transformer for Universal Image Segmentation (Mask2Former),” *CVPR*, 2022.
- [5] M. Oquab *et al.*, “DINOv2: Learning Robust Visual Features without Supervision,” *TMLR*, 2024.
- [6] T. Keressies *et al.*, “Your ViT is Secretly an Image Segmentation Model,” *CVPR*, 2025.
- [7] N. Nayal *et al.*, “RbA: Segmenting Unknown Regions Rejected by All,” *ICCV*, 2023.
- [8] D. Hendrycks *et al.*, “Scaling Out-of-Distribution Detection for Real-World Settings,” *ICML*, 2022.
- [9] E. Hu *et al.*, “LoRA: Low-Rank Adaptation of Large Language Models,” *ICLR*, 2022.
- [10] E. Romera *et al.*, “ERFNet: Efficient Residual Factorized ConvNet for Real-Time Semantic Segmentation,” *IEEE T-ITS*, 2018.
- [11] A. Kirillov *et al.*, “Panoptic Segmentation,” *CVPR*, 2019.
- [12] M. Cordts *et al.*, “The Cityscapes Dataset for Semantic Urban Scene Understanding,” *CVPR*, 2016.
- [13] T.-Y. Lin *et al.*, “Microsoft COCO: Common Objects in Context,” *ECCV*, 2014.
- [14] D. Hendrycks, K. Gimpel, “A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks,” *ICLR*, 2017.